

Assignment on

“Predicting Taxi Trip Fares Using Machine Learning”

WM-ASDS04: Introduction to Data Science with Python (Section A)

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1. Introduction

1.1 Background

The taxi industry is a cornerstone of urban transportation, facilitating millions of trips daily across cities worldwide. Pricing in the taxi industry is a complex function of multiple interacting factors, including trip distance, time of day, traffic conditions, weather, and passenger demand. Traditionally, taxi fare structures have been determined by regulatory bodies using fixed per-kilometre and per-minute rates, often supplemented by base fares and dynamic surge pricing during peak demand periods.

With the rapid growth of ride-hailing platforms such as Uber, Lyft, and Pathao — the latter being a popular option in Bangladesh — data-driven fare prediction has become an essential tool for both service providers and passengers. Accurate fare prediction improves customer trust and satisfaction, helps drivers plan their schedules more effectively, and allows operators to optimise pricing strategies in response to real-time conditions.

Machine learning offers a powerful approach to fare prediction by learning patterns from historical trip data. Unlike rule-based pricing systems, machine learning models can capture complex, non-linear relationships between input features and the target fare, potentially improving prediction accuracy significantly. This study applies two machine learning models — Linear Regression and Random Forest Regression — to a synthetic taxi trip dataset and evaluates their comparative performance.

1.2 Objectives

The primary objectives of this study are as follows:

- To explore and preprocess a real-world taxi trip dataset, addressing missing values and encoding categorical variables appropriately.
- To perform Exploratory Data Analysis (EDA) to uncover patterns and relationships within the dataset.
- To build and train two predictive regression models: Linear Regression and Random Forest Regression.
- To evaluate and compare model performance using standard metrics: R^2 , Mean Absolute Error (MAE), and Root Mean Square Error (RMSE).
- To identify the most significant features contributing to taxi fare prediction.
- To draw meaningful conclusions and insights applicable to the taxi and ride-hailing industry.

2. Methodology

2.1 Data Collection

The dataset used in this study is a synthetic taxi trip dataset sourced from Kaggle, published under the title "Taxi Price Prediction" by the user denkuznetz. The dataset was specifically designed for regression tasks, providing realistic simulated data for the purpose of exploring pricing trends in the taxi industry. Although synthetic in nature, the dataset reflects the key variables that influence real-world taxi fares, making it appropriate for this predictive modelling exercise.

The dataset contains 1,000 records and 11 features covering trip characteristics, environmental conditions, and fare components. Table 1 below provides a detailed description of each feature in the dataset.

Table 1: Dataset Feature Description

Feature	Data Type	Description
Trip_Distance_km	float64	Distance of the trip in kilometres
Time_of_Day	object	Time period of the trip (Morning, Afternoon, Evening, Night)
Day_of_Week	object	Day of the week (Monday through Sunday)
Passenger_Count	float64	Number of passengers in the trip
Traffic_Conditions	object	Traffic level (Low, Medium, High)
Weather	object	Weather condition during the trip
Base_Fare	float64	Base fare charged for the trip
Per_Km_Rate	float64	Fare charged per kilometre
Per_Minute_Rate	float64	Fare charged per minute of travel
Trip_Duration_Minutes	float64	Total duration of the trip in minutes
Trip_Price	float64	Target variable — total price of the trip

2.2 Data Preprocessing

Prior to modelling, the raw dataset underwent a series of preprocessing steps to ensure data quality and compatibility with machine learning algorithms.

Missing Value Treatment: The dataset contained 50 missing values distributed across 10 of the 11 columns. The target variable Trip_Price had only 1 missing value, whereas all other columns had exactly 50 missing values each. Numerical columns were imputed using the median value to minimise the influence of outliers, while categorical columns were imputed using the mode (most frequent value) to preserve the distributional characteristics of each category.

Categorical Encoding: Four categorical features — Time_of_Day, Day_of_Week, Traffic_Conditions, and Weather — were encoded using Label Encoding, converting each category into a numeric integer representation. This step was necessary as machine learning algorithms require numeric input.

Feature Selection: All 10 available features (excluding the target variable Trip_Price) were included as input features for model training, ensuring the models had access to the full set of available information.

Train-Test Split: The dataset was split into a training set of 800 samples (80%) and a test set of 200 samples (20%) using a fixed random state of 42 to ensure reproducibility.

Feature Scaling: StandardScaler was applied to normalise the features for the Linear Regression model, transforming each feature to have zero mean and unit variance. The Random Forest model, being a tree-based algorithm, does not require feature scaling and was trained on the unscaled data.

2.3 Analysis Techniques

This study employed two distinct regression techniques, chosen to represent a spectrum of model complexity:

Linear Regression: A parametric, interpretable model that assumes a linear relationship between the input features and the target variable. It serves as the baseline model in this study, providing a benchmark against which the performance of the more complex model can be assessed.

Random Forest Regression: An ensemble learning method that constructs multiple decision trees during training and outputs the mean prediction of the individual trees. Random Forest is capable of capturing non-linear relationships and interactions between features, making it well-suited to complex datasets. The model was configured with 100 estimators and a random state of 42.

Model performance was evaluated using three standard regression metrics: R^2 (coefficient of determination), which measures the proportion of variance in the target variable explained by the model; Mean Absolute Error (MAE), which measures the average magnitude of prediction errors; and Root Mean Square Error (RMSE), which penalises larger errors more heavily than MAE.

2.4 Technologies and Tools

The entire analysis was conducted in Python, leveraging the following key libraries:

- pandas — for data loading, cleaning, and manipulation.
- NumPy — for numerical computation and array operations.
- Matplotlib and Seaborn — for data visualisation and exploratory analysis.
- scikit-learn — for machine learning modelling, preprocessing, and evaluation metrics.

All code was written and executed in a standard Python environment. The dataset was stored locally as a CSV file and loaded using `pandas.read_csv()`.

3. Case Study / Practical Application

3.1 Application of Data Science to Taxi Fare Prediction

This case study applies data science techniques to the problem of predicting taxi trip fares based on a combination of trip characteristics and environmental factors. The goal is to demonstrate how a machine learning pipeline can be built end-to-end — from raw data ingestion to model evaluation — using real-world applicable methodology.

The taxi fare prediction problem is framed as a supervised regression task, where the model learns the relationship between trip features and the target fare from the training data, and then generalises this knowledge to predict fares on unseen data.

3.2 Exploratory Data Analysis

The EDA phase revealed several important patterns in the dataset. The distribution of Trip_Price showed a roughly right-skewed distribution, with most fares concentrated in the lower to mid-range and a tail of higher-priced trips. The correlation heatmap demonstrated that Trip_Distance_km, Trip_Duration_Minutes, Base_Fare, Per_Km_Rate, and Per_Minute_Rate were the most strongly correlated features with the target variable Trip_Price. Traffic conditions were also found to influence fare levels, with high-traffic trips generally associated with higher fares due to increased trip durations.

Advanced visualisations produced during the EDA phase included a correlation heatmap showing pairwise relationships across all numeric features, and a scatter plot of Trip Distance versus Trip Price colour-coded by traffic condition, which clearly illustrated how higher traffic density was associated with disproportionately higher fares for equivalent distances.

3.3 Model Results and Comparison

Both models were trained and evaluated on the same train-test split. Table 2 below summarises the performance metrics for each model on the test set.

Table 2: Model Performance Comparison

Model	R ²	MAE	RMSE	Performance
Linear Regression	0.7700	9.8423	13.8223	Baseline
Random Forest	0.9288	5.4137	7.6913	Best Model ✓

Linear Regression achieved an R² of 0.7700, a MAE of 9.8423, and an RMSE of 13.8223. While this indicates that approximately 77% of the variance in fare prices is explained by the model, the relatively high MAE and RMSE suggest that linear relationships alone are insufficient to fully capture the complexity of fare pricing.

Random Forest Regression significantly outperformed Linear Regression, achieving an R² of 0.9288, a MAE of 5.4137, and an RMSE of 7.6913. This means the Random Forest model explains approximately 93% of fare variance, with prediction errors that are substantially lower than the baseline model. The improvement in MAE (from 9.84 to 5.41) represents a 45%

reduction in average prediction error, which would translate to meaningfully better fare estimates in a real-world deployment scenario.

Feature importance analysis from the Random Forest model further highlighted that `Per_Km_Rate`, `Trip_Distance_km`, and `Base_Fare` were the three most influential predictors of taxi fare, which aligns with domain knowledge about how taxi pricing is structured.

3.4 Discussion of Outcomes

The results confirm that ensemble methods such as Random Forest are significantly better suited to taxi fare prediction than linear models. The non-linear interactions between distance, duration, traffic conditions, and pricing rates are effectively captured by the tree-based ensemble approach but not by the linear model.

In a practical context, a fare prediction model with an R^2 of 0.9288 could be deployed in a ride-hailing application to provide passengers with accurate fare estimates before they book a trip. This builds trust, reduces fare disputes, and improves the overall user experience. For operators, such a model enables dynamic pricing optimisation and revenue forecasting.

The synthetic nature of the dataset does impose some limitations. In a real-world deployment, additional features such as GPS coordinates, driver ratings, vehicle type, and seasonal demand patterns would likely improve predictive accuracy further. Future work could also explore gradient boosting methods such as XGBoost or LightGBM, as well as deep learning approaches for even higher performance.

4. Conclusion

This study successfully demonstrated the application of machine learning techniques to the problem of taxi fare prediction. Starting from a synthetic dataset of 1,000 taxi trips, a complete data science pipeline was constructed encompassing data loading, missing value imputation, categorical encoding, feature scaling, exploratory data analysis, model training, and performance evaluation.

Two models were trained and compared: Linear Regression, serving as the interpretable baseline model, and Random Forest Regression, an ensemble learning approach. The results clearly indicated the superiority of the Random Forest model, which achieved an R^2 score of 0.9288 compared to 0.7700 for Linear Regression, along with substantially lower MAE and RMSE values. All stated objectives were successfully achieved.

The analysis confirmed that trip distance, per-kilometre rate, and base fare are the primary drivers of taxi trip price, consistent with real-world pricing logic. The visualisations further illuminated the influence of traffic conditions and trip duration on fare variability.

In conclusion, machine learning — particularly ensemble-based methods — offers a robust and accurate approach to fare prediction in the taxi and ride-hailing domain. The methodology and findings presented in this study provide a solid foundation for more advanced predictive modelling work, including real-time fare estimation systems and dynamic pricing engines.

5. References

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Work Distribution

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